

# Senudi Rupasinghe

Business & Technical Analysis · Full-Stack Software Development

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## SUMMARY

Information Technology graduate (SLIIT, 2026) who bridges business needs and technical solutions. Spent six months at WSO2 turning the sales team's needs into working features across a React/TypeScript frontend and Ballerina backend integrations. Final-year research delivered a fully documented, IEEE-conference-published AI screening system (97.8% accuracy) — requirements analysis through solution specification, testing and evaluation. Strong in SQL, data architecture and clear technical documentation.

## EXPERIENCE

### Software Engineering Intern (Digital Transformation) - WSO2

Jan 2025 – Jun 2025

*Sales Pitstop — internal application used by WSO2's global sales team*

- Translated the sales team's business needs into working product features, building and maintaining the frontend in React, TypeScript and Redux.
- Developed backend integration logic in Ballerina (WSO2's integration language), connecting the application to internal systems.
- Worked in an Agile team with Git branch-and-PR workflows, peer code reviews and iterative delivery.

## PROJECTS

### Smart-Agri-Suite - Final-Year Research Project (Individual Component)

2025 – 2026

*React Native · TypeScript · FastAPI · Python · scikit-learn · Agora RTC*

- Cultivator Intention Analyzer (individual thesis): analysed the problem domain, specified requirements, and designed a multi-modal behavioral-screening pipeline - an intent-risk classifier reaching 97.8% test accuracy over 16 fused audio + text features on 1,155 labelled samples.
- Engineered an auditable, risk-aware decision architecture: the LLM (Groq / Llama-3.3-70B) is constrained to explanation generation only, and degraded or unknown evidence never auto-approves.
- Produced full research documentation - requirements, solution specification and experimental evaluation - reviewed by academic panels.

### StudyMate - RAG Study Companion

In progress, 2026

*React · TypeScript · FastAPI · Python · PostgreSQL + pgvector · LangChain. OpenAI API*

- Building and deploying a retrieval-augmented study tool where students upload lecture PDFs and get source-grounded answers; designed the data architecture around PostgreSQL with pgvector embeddings.

## PUBLICATIONS

**"Smart Agri-Suite: A Globally Deployable Machine Learning Platform for Satellite-Driven Idle Land Mobilization and Smallholder Agricultural Decision Support"** - co-author; peer-reviewed paper published in the proceedings of the 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA 2026), IEEE, May 2026. DOI: 10.1109/ICHORA69329.2026.11537179

## SKILLS

**Business analysis:** Requirements gathering & translation, technical documentation, solution specification, Agile, stakeholder collaboration

**Data & architecture:** SQL, PostgreSQL, MongoDB, SQL Server, data modelling, REST API design

**Engineering:** TypeScript, JavaScript, Python, Java, C#; React, React Native, Redux, FastAPI, Node.js/Express, Ballerina

**AI / ML:** LLM APIs (Gemini, Groq), retrieval-augmented generation, scikit-learn

## EDUCATION

**BSc (Hons) in Information Technology - SLIIT, Malabe**

2022 – 2026

**GCE Advanced Level - Lyceum International School, Wattala**

2020 – 2022

*Mathematics, Computer Science, Physics*

## ADDITIONAL

**Languages:** English (full professional), Sinhala (limited working) · **Activities:** AIESEC; School Colours - Swimming